

Why Right Intentions Are Not Enough for Artificial Moral Agents

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As the consequences of our actions are unknowable in advance, thus for an artificial agent, it is good enough that it formulates the right intentions to act. That we could call a “good” agent.

The purpose of this essay is to discuss, refute, and provide counterarguments to the argument above. There are two parts to this argument: a premise and a conclusion. I will begin by refuting the premise in the context of artificial agents. In the case of a false premise, the argument becomes logically unsound. Then, I would assume that the given premise is true and show why it does not necessarily lead to the conclusion.

Let’s begin with the premise. The premise of the argument, “all consequences are unknowable in advance”, does not hold water, as some consequences of our actions are clearly foreseeable. For example, if you put your finger on fire, it will burn. The same logic applies to artificial agents. In the game of tic-tac-toe, a robot could easily exhaustively test all possible moves and positions using a search tree and choose the optimal moves (Fox, 2013). In this case, the robot player could foresee all possible outcomes and act with its intention to win the game.

Even if we are faced with a complex situation with many uncertainties, as human beings, we can still reason about the consequences based on knowledge and experience. For example, we know the chances of getting into an accident are higher if people ride their motorbikes faster, even though we do not know the traffic on every road. Similarly, probabilistic reasoning of consequences is an essential part of the design of many AI applications (Jeevanandam, 2017). For example, probability theory and Bayesian inference are at the core of many machine learning algorithms for prediction (Brownlee, 2019a; Torfi, 2020). A language translation app, for instance, minimises the matching error in an optimisation problem to find the most probable translation (Németh, 2019). In this case, artificial agents take a probabilistic view of all possible outcomes (i.e. translations) and tailor their actions to maximise the chance of achieving the best results. Even in cases where the unknown distribution of consequences is computationally intractable, approximate methods such as Markov chain Monte Carlo can still provide insights by drawing samples from the unknown distribution (Brownlee, 2019b). In utilitarian terms, since achieving the optimal outcome maximises its utility, it could be considered its intention.

While probability inference gives some artificial agents the ability to reason probabilistically, another type of agent, the reinforcement learning (RL) agent, learns the consequences of actions through experience by interacting with an unknown environment (Osiński and Budek, 2018). An RL agent works as follows: given a set of permitted actions and states, the agent seeks a specified reward and experiments with combinations of sampled state-action pairs to maximise it. In an iterative trial-and-error process, the agent learns the consequences of actions, navigates the unknown environment, and pursues a series of locally optimal strategies that maximise rewards (Suh, 2021). In this process, understanding previous consequences (e.g. which move gives the greatest reward) strengthens the agent’s subsequent actions and eventually leads to a better understanding of all possible consequences.

As discussed so far, artificial agents can know the consequences of their actions to a great extent through exhaustive searches, probabilistic inferences, and reinforced learning. Therefore, the premise of the argument is not valid. Now let us take a twist and assume that the premise of this argument

is actually valid, i.e. the consequences of artificial agents’ actions are unknowable in advance. This is true in some cases. For example, self-driving cars are programmed to avoid collisions with other vehicles, but they may cause accidents if their engines malfunction unexpectedly. Having established that the premise is true, one must determine if formulating the “right intentions” is “good” enough for artificial agents. To start with, it is important to define what “good” and “right intentions” mean in the context. In the following paragraphs, I will discuss these two concepts from (i) a utilitarian, and (ii) a Kantian deontological perspective.

Suppose we adopt utilitarians’ view that the “right intention” to act is to maximise utility (Driver, 2022). A probable consequence utilitarian (PCU) and an actual consequence utilitarian (ACU) would have slightly different views about what utility should be maximised (Strasser, 1989). For PCUs, we are morally “good” if we choose the action that maximises the expected utility out of all possible actions, even if the actual outcome is much worse. In other words, the “right intention” is to choose the action that maximises the foreseeable, but not the actual, benefits. This reflects their deep recognition of uncertainty in consequences, as recognised in the premise, where unforeseeable incidents can result in unintended consequences for actions.

However, ACUs may argue that, despite being “rational” to choose an action that maximises expected utility, it may not necessarily be “morally good” if it causes disastrous consequences. For example, many social media companies use complex machine learning algorithms to personalise content for users, with the aim to maximise their engagement (Boyd, 2010). However, they often ended up promoting ethically-questionable or polarised posts that evoke strong emotions as those tend to get more views (DeLeon, 2019). In this case, ACUs would argue that the company’s actions are morally wrong following Bentham’s “total happiness” principle, because they harm the public more than they benefit it, even though the company’s initial intention is merely to maximise expected revenue.

Having examined the utilitarian views on “right intentions” and “good” actions for artificial agents, my next step is to investigate the Kantian deontology viewpoint. According to Kant’s ethics, the right intention is to follow the categorical imperative, that is, act according to maxims or rules that one believes generalise to universal laws (Johnson and Cureton, 2022). In order to create such maxims for artificial agents, we must take into account the type of AI systems we are working on. When the AI system is fully autonomous, it can produce its own rules without consulting humans (Walch, 2020). If the system is human-machine integrated, then rules should be created by a combination of human and machine interaction and intervention to aid human-centered tasks (Wilson and Daugherty, 2018). In the next paragraph, I will discuss the challenges of defining universal maxims that produce morally good outcomes for both systems.

A fully autonomous AI system makes its own decisions and rules based on its own database and past experiences without human intervention (Walch, 2020). However, due to the inherent black-box nature of complex AI systems, this rule-generation process is highly uncertain, and it may be restricted due to a lack of data and learning experiences (Ulgen, 2017). The unpredictable nature of rule generation may have detrimental effects on humanity. A fully autonomous weapon system, which selects and attacks human targets based on previous military scenarios, may kill innocent people to crush the enemies (Stauffer, 2020), thus fundamentally violating Kant’s ethical principles. As with Kant’s “hypothetical imperatives” (Johnson and Cureton, 2022), here it is the machine subjectivity that dictates the actions, which are motivated by subjective desires that aren’t always good. A human-machine integrated system, on the other hand, may encounter problems when the interface between humans and machines fails, such as system failures and communication breakdowns, even though humans are at the center of decision-making (Ulgen, 2017). This may prove fatal in integrated human-weapon systems that rely heavily on communication and coordination. In this situation, even if the agent desires to follow and execute the right intention, it is incapable of doing so, thus leading to undesirable results. Therefore, while both systems could be morally justified in adhering to good maxims, it may not even be possible for them to formulate universally good maxims in the first place, or to practically follow them.

To summarise, I have provided three examples that refute the premise of the argument, showing that artificial agents can at least know the consequences of their actions to some extent. I then at-

tempted to demonstrate that, even assuming a true premise, formulating the right intentions may not be good enough in the eye of (i) actual consequence utilitarians, because following them may result in unexpected catastrophic consequences; (ii) Kantian deontology, since it may be impossible to formulate the right intentions at all, and the inability to follow those intentions may also result in bad outcomes. As the premise cannot lead to the conclusion, the argument is not valid.

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